

Layer 2 Device Behavior Analysis

Network Architecture Comparison: Hubs vs Switches



The Scenario

TechCorp Industries operates a distributed office environment with four functional departments: Finance, Human Resources, Information Technology, and Marketing. Over the past fiscal quarter, the organization has experienced progressive degradation in network performance during peak operational windows (typically 10:00 AM to 3:00 PM). The Finance department has reported inconsistent file transfer speeds when conducting large data transactions, particularly when other departments are simultaneously active on the network. System administrators have noted frequent collision warnings in network logs. The Chief Technology Officer suspects that the existing network infrastructure—currently based on legacy hub technology—may be the limiting factor constraining overall system throughput. Before authorizing capital expenditure for network equipment upgrades, executive management has requested empirical evidence demonstrating the performance and behavioral differences between the current hub-based architecture and a proposed switch-based alternative.



Guidelines

Experimental Framework

Construct two identical network topologies in separate Cisco Packet Tracer projects. Each topology shall consist of one central Layer 2 device, four personal computers, and copper straight-through cables for interconnection.

Network Configuration

Department	Device	Address
Finance	PC-Finance	192.168.1.10
Human Resources	PC-HR	192.168.1.20
Information Technology	PC-IT	192.168.1.30
Marketing	PC-Marketing	192.168.1.40

Testing Scenarios

Activate Simulation Mode in Cisco Packet Tracer. Execute these test scenarios in both topologies:

- Test 1:** *Single Unicast:* Send a data packet from Finance to Human Resources and observe port activity
- Test 2:** *Concurrent Transmission:* Simultaneously send packets from Finance to Human Resources and from Marketing to Information Technology; record collision behavior
- Test 3:** *Broadcast:* Send a broadcast packet to 192.168.1.255 and forwarding paths

 Submission**Deliverables**

Submit the following as a single PDF document:

1. Both Cisco Packet Tracer project files (HubNetwork.pkt and SwitchNetwork.pkt)
2. Annotated screenshots showing topology and packet flow for both devices across all test scenarios
3. Port activity and packet path comparison between the hub and switch